UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Dipartimento
di Fisica
e Astronomia
Galileo Galilei

No U-Turn Sampler

Adaptively Setting Path Lengths in Hamiltonian Monte Carlo

Pieripolli Leonardo, Pirazzo Tommaso,
Ponchio Mattia, Secco Benedetto

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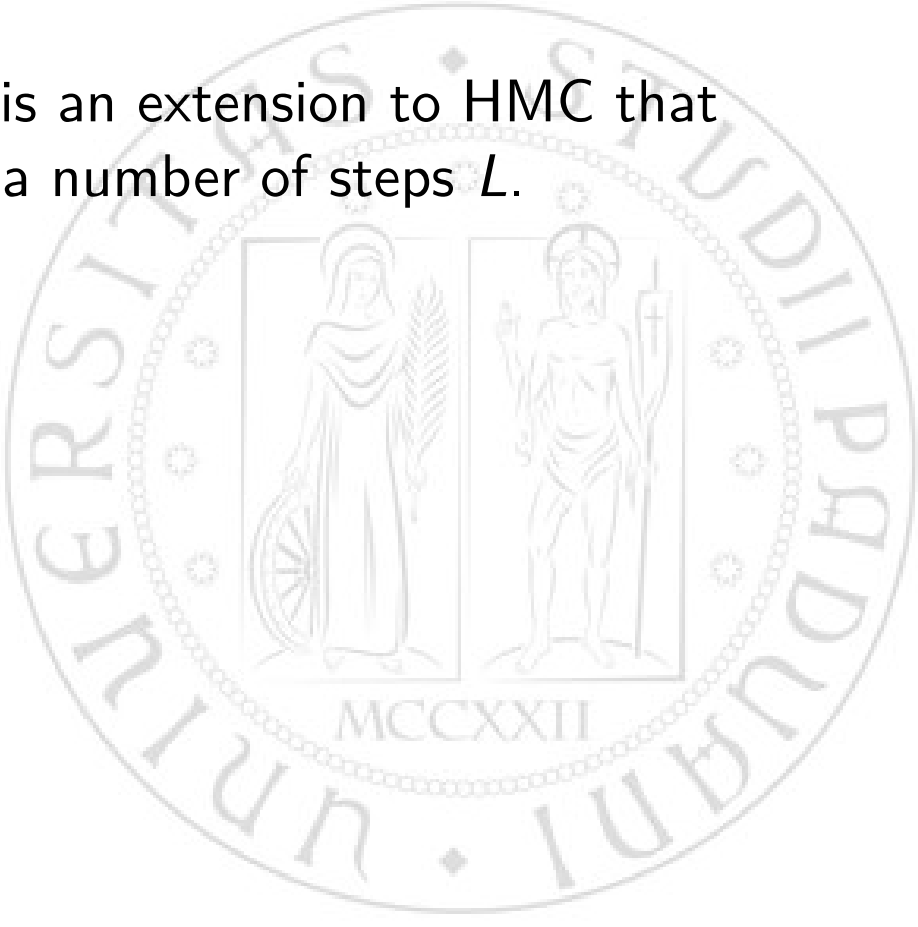

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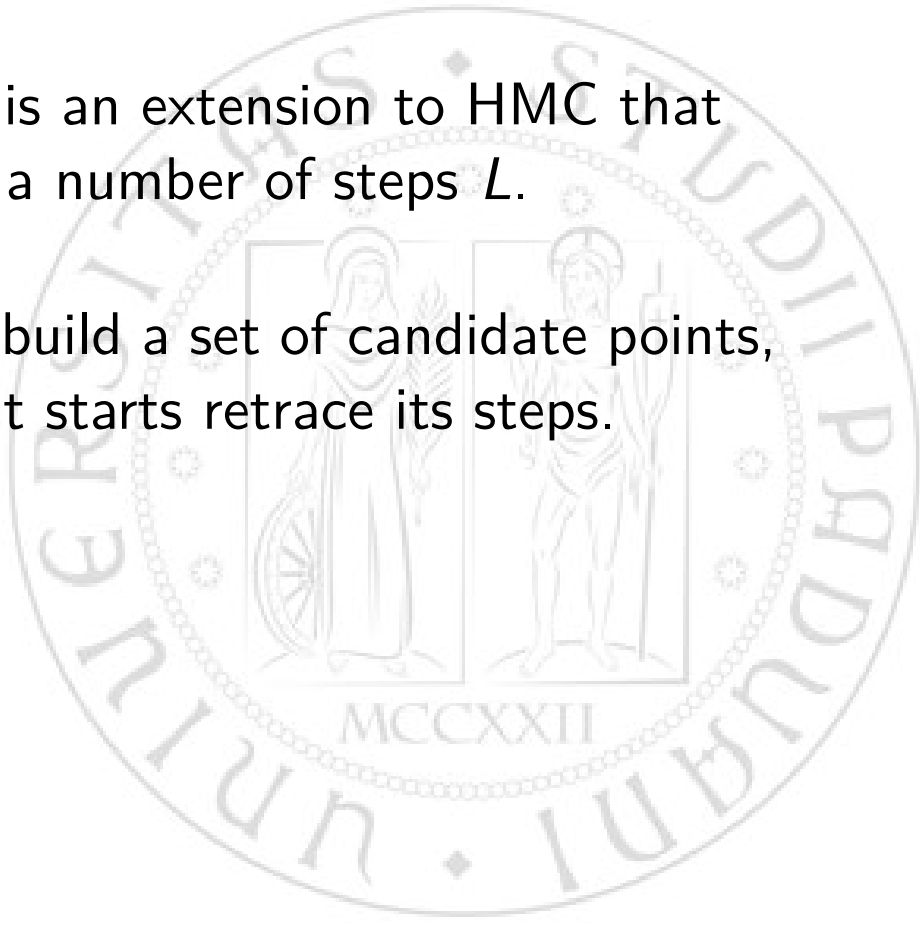

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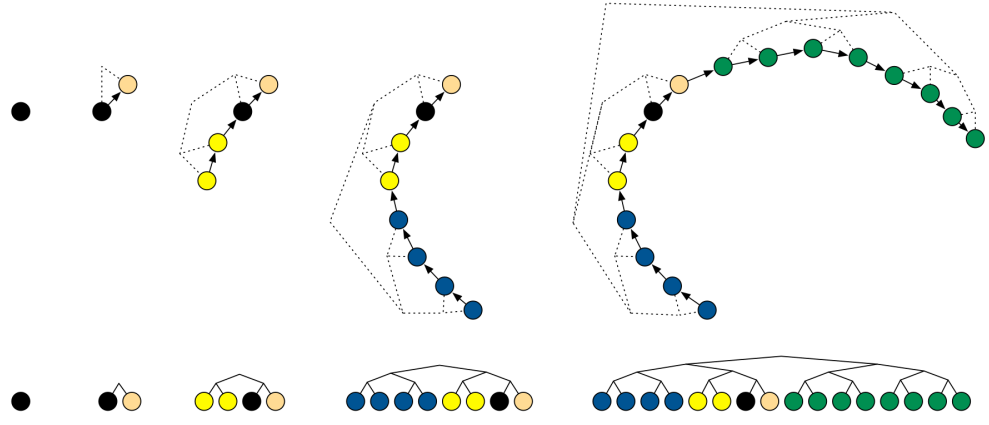

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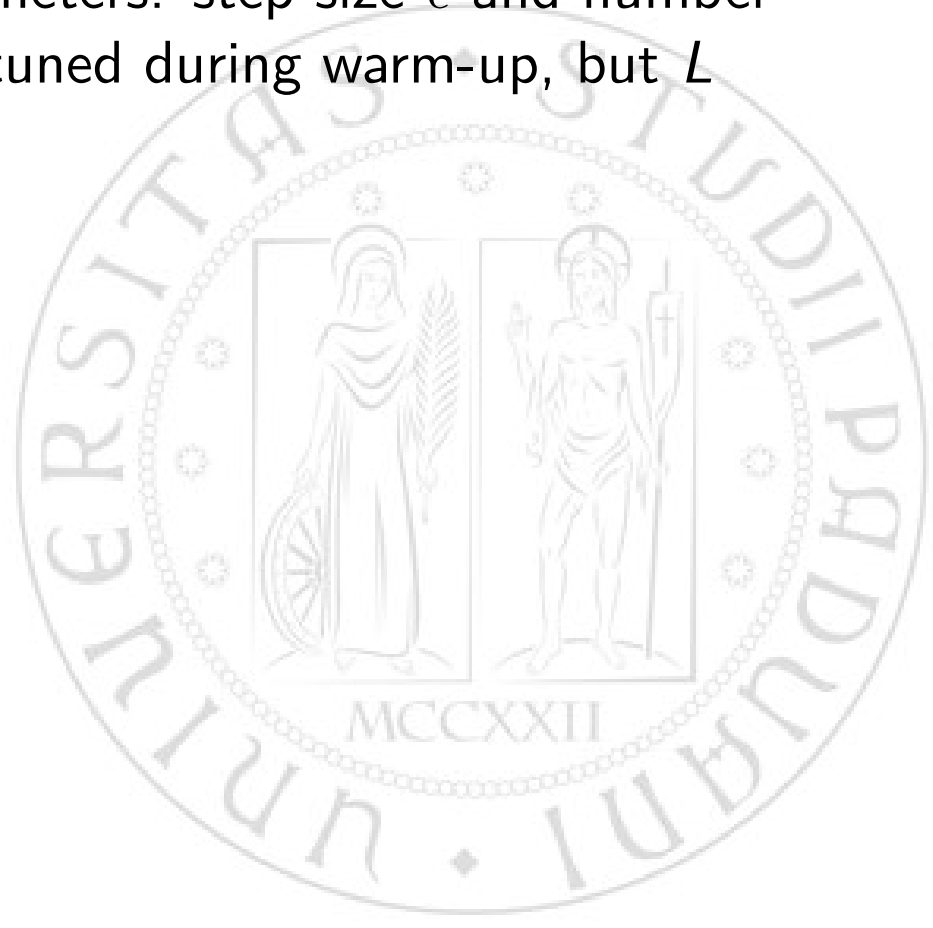

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Algorithm 1 Hamiltonian Monte Carlo

Given θ^0, ϵ, L, $\mathcal{L}$, M:
for $m = 1$ to M **do**
  Sample $r^0 \sim \mathcal{N}(0, I)$.
  Set $\theta^m \leftarrow \theta^{m-1}, \tilde{\theta} \leftarrow \theta^{m-1}, \tilde{r} \leftarrow r^0$.
  for $i = 1$ to L **do**
    Set $\tilde{\theta}, \tilde{r} \leftarrow \text{Leapfrog}(\tilde{\theta}, \tilde{r}, \epsilon)$.
  end for
  With probability $\alpha = \min\left\{1, \frac{\exp\{\mathcal{L}(\tilde{\theta}) - \frac{1}{2}\tilde{r}\cdot\tilde{r}\}}{\exp\{\mathcal{L}(\theta^{m-1}) - \frac{1}{2}r^0\cdot r^0\}}\right\}$, set $\theta^m \leftarrow \tilde{\theta}$, $r^m \leftarrow -\tilde{r}$.
end for

function Leapfrog(θ, r, ϵ)
Set $\tilde{r} \leftarrow r + (\epsilon/2)\nabla_\theta \mathcal{L}(\theta)$.
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return $\tilde{\theta}, \tilde{r}$.

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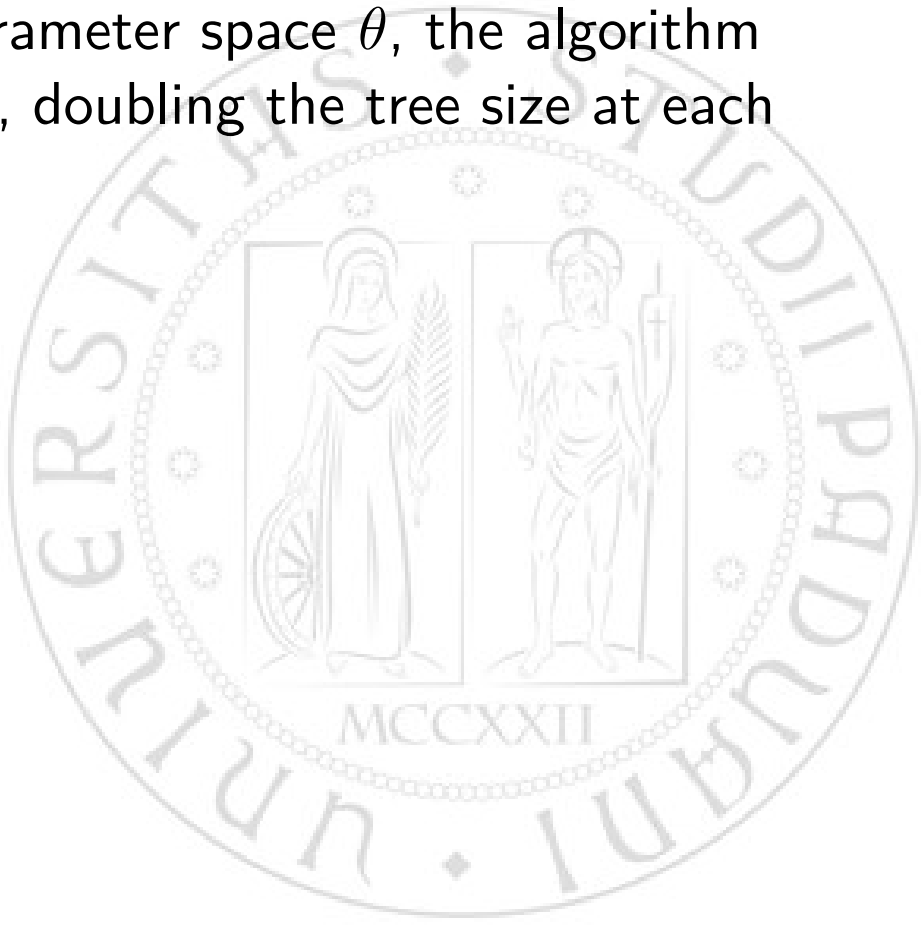
- Starting from a point in the parameter space θ, the algorithm builds a set of candidate points, doubling the tree size at each iteration.



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- Starting from a point in the parameter space θ, the algorithm builds a set of candidate points, doubling the tree size at each iteration.
- To detect when to stop, the algorithm checks if the trajectory starts to turn back on itself: given a proposed point $\tilde{\theta}$, the algorithm checks if the dot product of the momentum vectors at θ and $\tilde{\theta}$ is negative.
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  $(\theta - \tilde{\theta}) \cdot \frac{d}{dt}(\theta - \tilde{\theta}) = (\theta - \tilde{\theta}) \cdot \tilde{r} < 0$
- To preserve time reversibility, NUTS runs the Hamiltonian dynamics in both forward and backward directions.

Probability slicing

To simplify the derivation and implementation of the algorithm, NUTS uses a slice variable u with conditional distribution

$$p(u|\theta, r) = \mathcal{U}(u; [0, exp\mathcal{L}(\theta) - \tfrac{1}{2} r \cdot r]).$$

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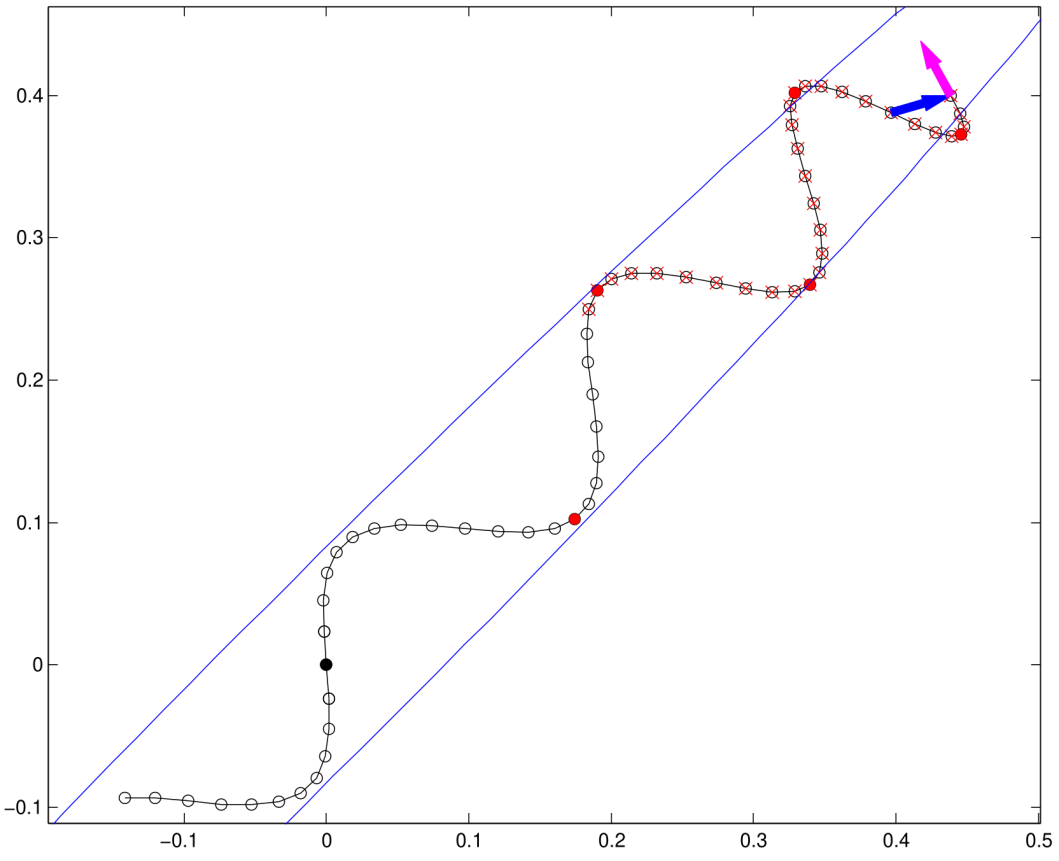
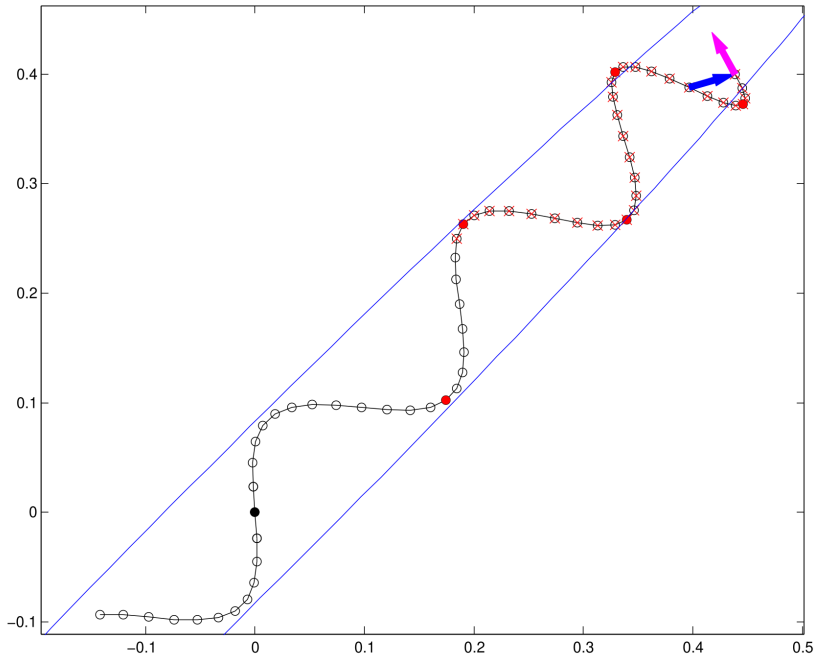


High-levl NUTS algorithm

- The algorithm resamples $u|\theta, r$ at each iteration,
- uses Leapfrog integrator to trace a path forward and backwards first 1 step,
- then doubling for successive iterations.
- This doubling process implicitly builds a balanced binary tree whose leaf nodes correspond to position-momentum states.
- The doubling is halted when the subtrajectory starts to double back on itself.



Example of a NUTS trajectory, with the slice variable u.